Cost of Inaction Analysis for Adolescent Girls' Education and Human Capital in Nepal

Hernando Grueso, PhD

UNICEF

October 2026

Section 1

Data and Context

Data Source: Nepal Living Standards Survey IV (NLSS IV) 2022/23

- **Source**: National Statistics Office of Nepal · NLSS IV 2022/23 microdata
- **Sampling**: Nationally representative two-stage stratified design across 7 provinces
- **Sample size**: 46,870 individuals in 9,600 households
- **Analysis sample**: 9,785 youth aged 15–24 (4,919 girls, 4,866 boys); 13,586 women aged 15–64 for the causal forest
- **Modules used**: S01 (roster), S07 (education), S08 (disability, Washington Group), S09 (employment), S10 (wages), poverty file
- **Survey design**: PSU = `psu_number`, strata = province, weights = `ind_wt`

Variables and Definitions

Concept	Definition	Source
NEET	Not in education, employment or training	S07 + S09
Enrolled	Currently attending any school level	q07_11
Secondary completed	Completed Class 10 (SLC) or higher	q07_12
Currently married	Self-reported marital status	q01_05
Caste/ethnicity	8 NLSS groups, recoded to 5 broad categories	q01_07
Wealth quintile	Asset-based, household level	constructed
Below poverty line	Official NSO poverty status	poverty file
Disability	Washington Group 6+ functional difficulty	q08_16–q08_21
Monthly wage	Long-term salaried workers, NPR	q10_07, q10_09

Section 2

Key Findings at a Glance

Five Headline Findings

1. Nepal's gender gap is **not in schools** — it is in labour markets. Girls' enrolment (40.2%) exceeds boys' (34.8%), yet NEET (46.8%) is nearly double boys' (26.4%).

2. **Marriage is the strongest driver of exclusion, and it begins in adolescence.** Being currently married is associated with a 61 pp reduction in enrolment and a 42 pp increase in NEET, dwarfing poverty, caste, and disability combined. The share married rises from 9% at ages 15–18 (where any current marriage is child marriage by definition) to 57% at 19–24.

3. **Secondary education has a robust estimated effect on inclusion**: a -13.2 pp reduction in NEET probability, stable across all provinces and caste groups (causal forest, n = 13,586).

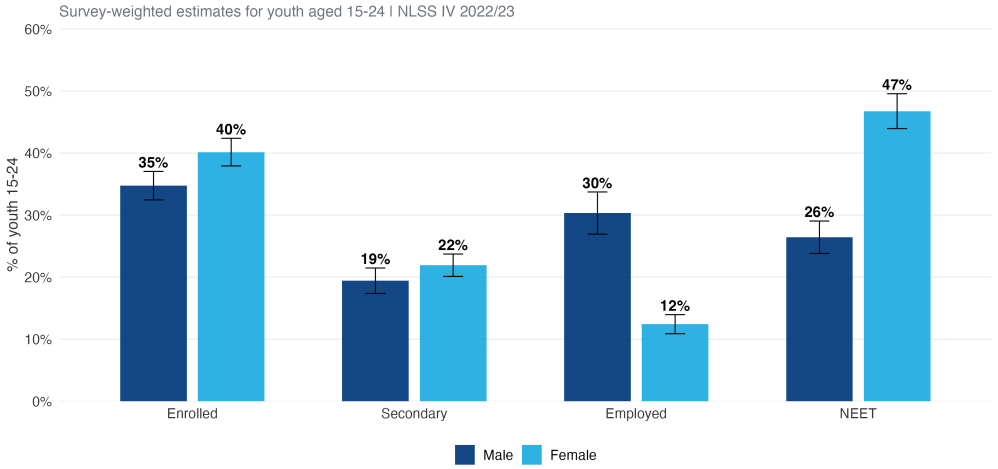
4. **Each additional year of schooling is associated with a 14.5% rise in lifetime earnings for girls vs 12.0% for boys.** Beyond secondary completion the gap *widens*: $\sim$+15% per year for girls vs $\sim$+10% per year for boys — the heart of the cost of inaction.

5. **Nepal forgoes $\sim$NPR 16 billion (USD 123 million) per cohort** by not educating its adolescent girls, equivalent to $\sim$**19% of the annual health-sector budget**, $\sim$15% of the social-security-allowance budget, and $\sim$8% of the education budget. Population counts are anchored to the 2021 Census.

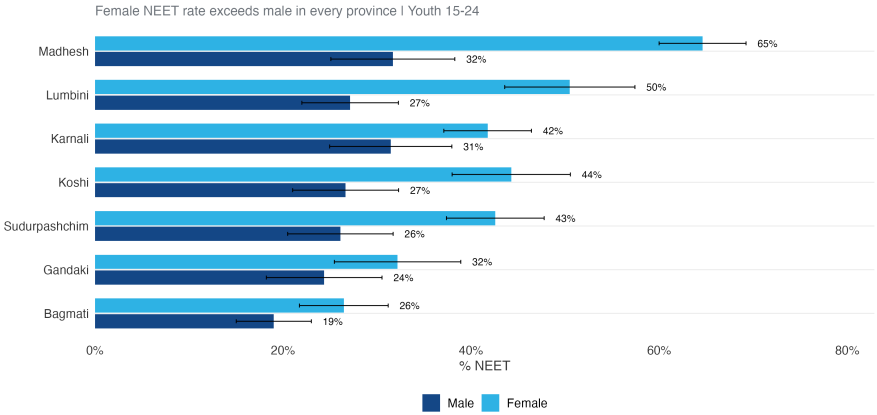
Section 3

Diagnostic: Gender Gaps in Education and Employment

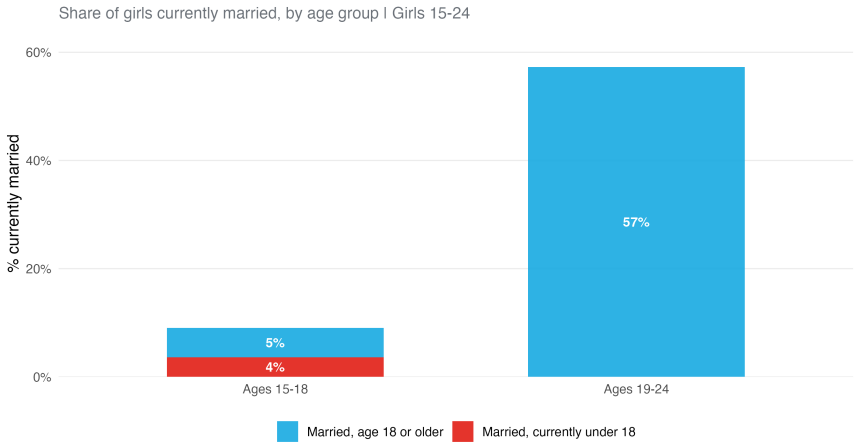
The Gender Gap is Not in Schools — It is in Labour Markets



Madhesh and Lumbini: Provinces with the Widest Female NEET Gap

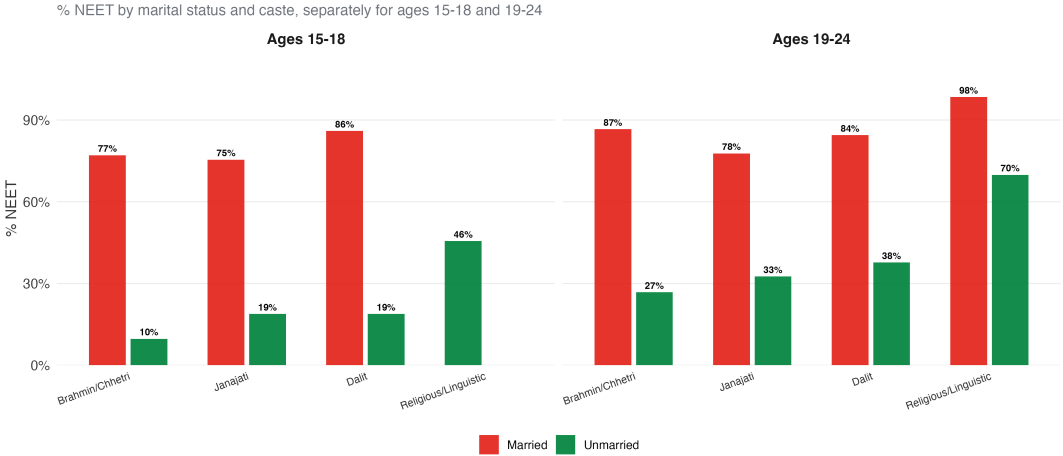


Marriage Rises Sharply Across the Adolescent Transition



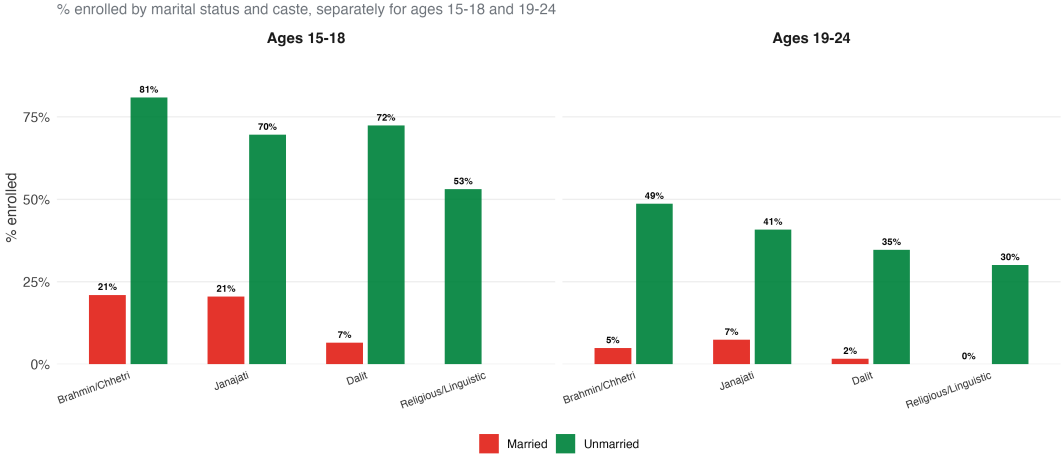
NLSS IV records current marital status, not age at first marriage. Among 15–18-year-olds any current marriage is child marriage by definition; for 19–24-year-olds only current status is observed.

The Marriage Penalty on NEET Holds in Both Age Groups, Regardless of Caste



NLSS IV 2022/23, survey-weighted. Cells with fewer than 15 observations dropped.

Married Girls' Enrolment Collapses in Both Age Groups, Across All Castes



NLSS IV 2022/23, survey-weighted. Cells with fewer than 15 observations dropped.

Headline Numbers — Where Each Girl Aged 15–24 Is

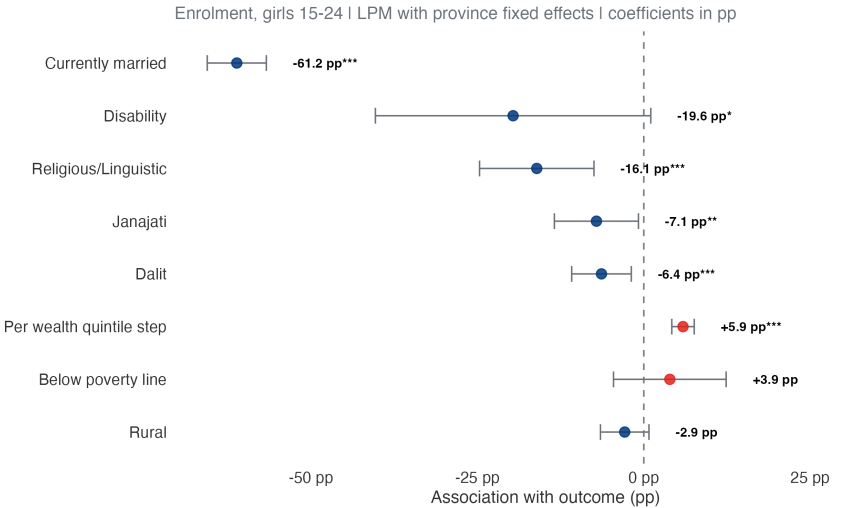
Status (youth 15–24)	Girls	Boys	Gap (pp)
In school only (not working)	35.4%	32.8%	+2.6 pp
In school **and** working	4.8%	2.0%	+2.8 pp
Working only (not in school)	13.0%	38.9%	**−25.9 pp**
NEET (Not in Educ/Employment)	**46.8%**	**26.4%**	**+20.4 pp**
Of which: currently married	37.4%	17.6%	+19.8 pp

The 46.8% NEET figure reflects girls who are *neither* in school *nor* employed under the ILO 7-day definition (which excludes unpaid family work and unsuccessful job search). These four categories are mutually exclusive and sum to 100%.

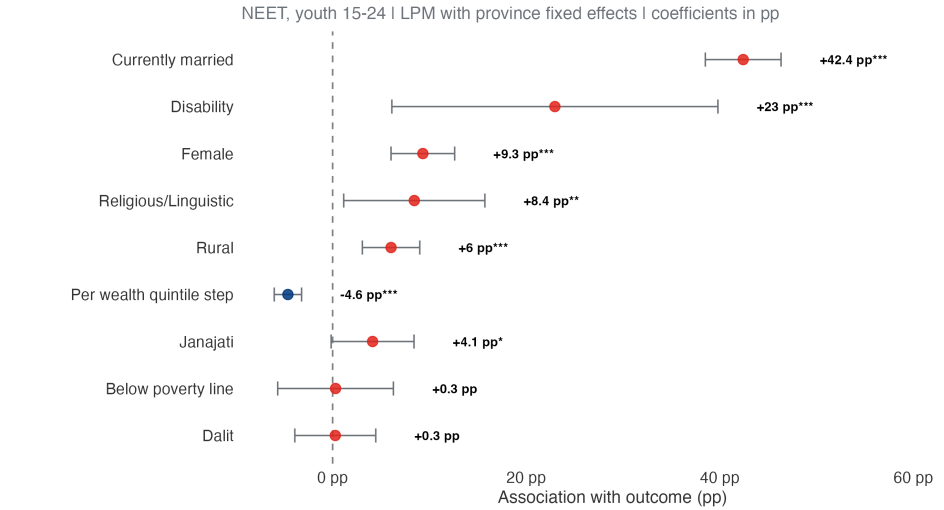
Section 4

Multivariate Analysis

What is Associated with Girls' Enrolment: All Factors in Percentage Points

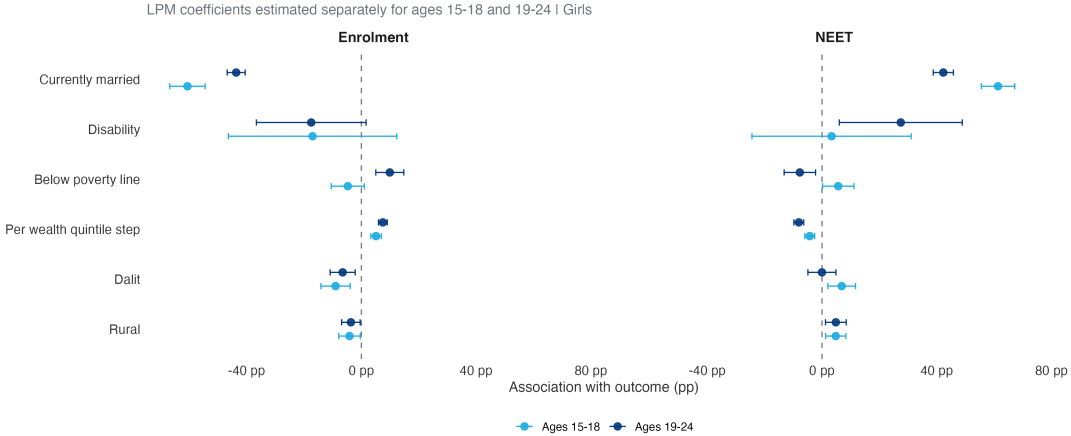


What is Associated with Youth NEET: All Factors in Percentage Points



NLSS IV 2022/23. Reference: Brahmin/Chhetri, urban, non-poor. *p<0.10 **p<0.05 ***p<0.01

The Marriage Association Strengthens with Age; Other Factors are Stable



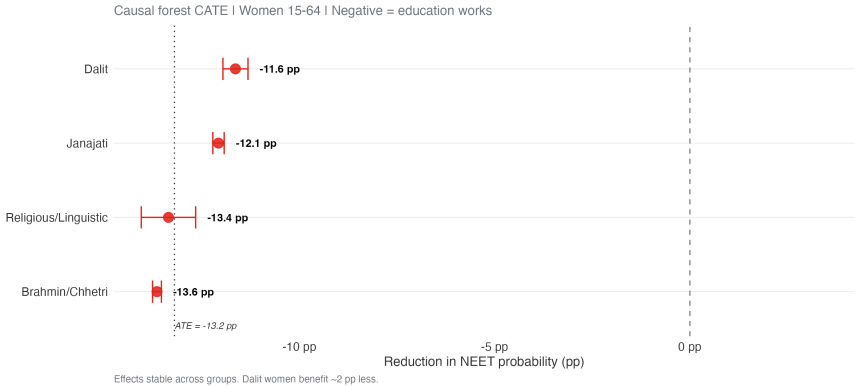
Reading the Coefficient Plot

- **Marriage is associated with the largest changes.** Being currently married is associated with a **−61 pp** lower enrolment probability and a **+42 pp** higher NEET probability, after controlling for poverty, caste, geography, and disability.
- **Caste matters even after wealth.** Dalit girls face a −6 pp enrolment gap that does not disappear when we control for wealth quintile. Religious/Linguistic girls face a similar penalty.
- **Wealth helps incrementally.** Each step up in the wealth quintile is associated with +5.9 pp enrolment and −4.6 pp NEET.
- **The gender coefficient on NEET remains +9.3 pp** even after holding everything constant — a residual labour-market gap not explained by observable characteristics.

Section 5

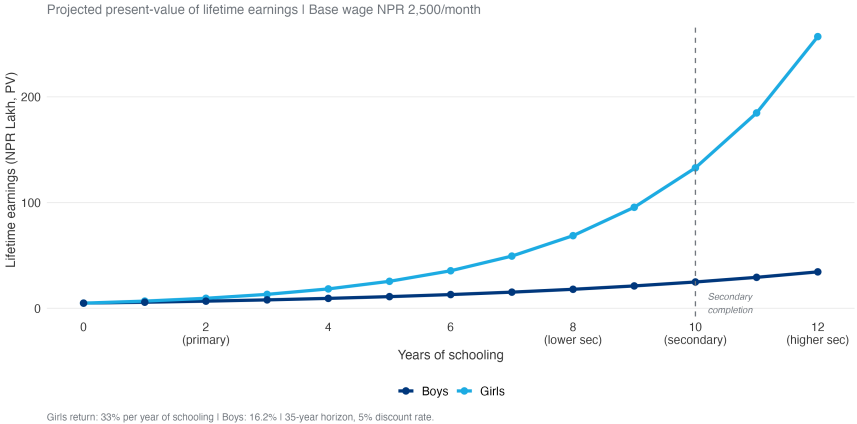
Estimating Education Returns for Adolescent Girls

Secondary Education Reduces NEET by ~13 pp — For Every Caste



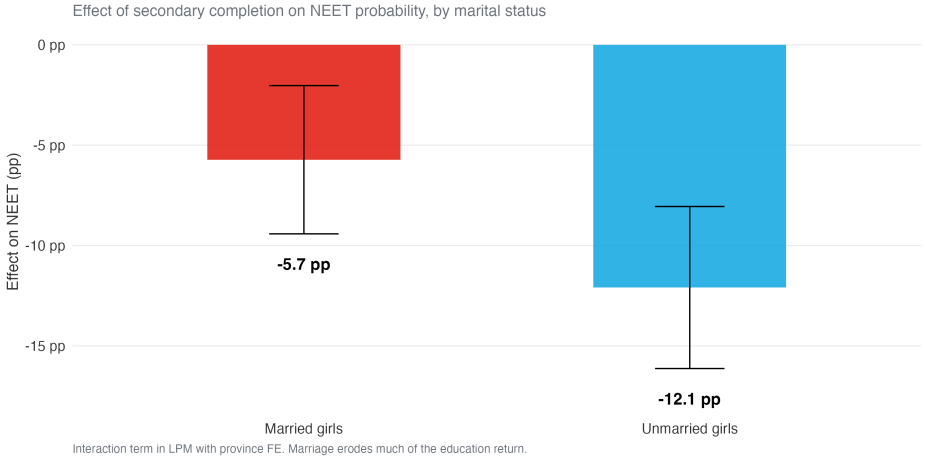
The −13.2 pp ATE estimate is for **women aged 15–64**. The parallel estimate for men is smaller (~−7 pp) — boys' NEET is shaped more by labour-demand conditions than by individual schooling. Causal-forest details and boys' coefficients in the appendix.

Returns to Schooling are Exponentially Larger for Girls



Average Mincer return: +14.5% per year of schooling for girls vs +12.0% for boys. **Beyond secondary** (10+ years): each additional year of schooling adds ~+15% to girls' lifetime earnings vs ~+10% for boys — the gender gap in returns *widens* with educational attainment. Compounded over 35 working years, this is where the cost of inaction concentrates.

Marriage Erodes the Protective Effect of Education



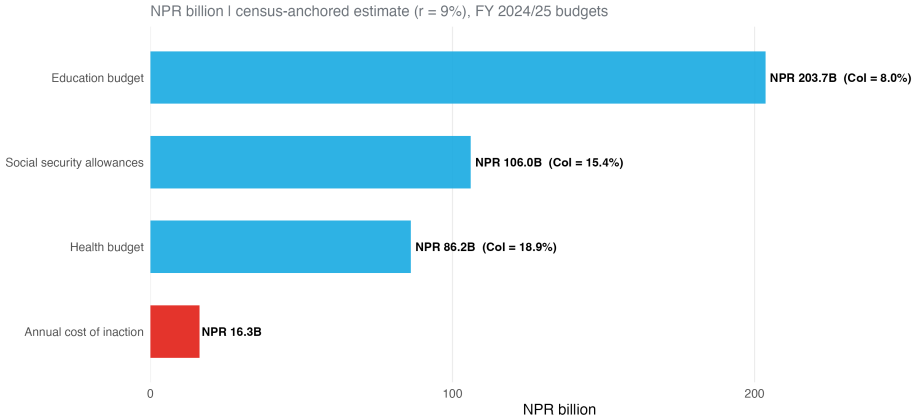
What the Analyses Tell Us

- **Education is associated with inclusion everywhere.** Completing secondary school is associated with a **−13.2 pp** lower NEET probability. The estimated effect is stable across all 7 provinces and all caste groups.
- **The Mincer return per year of schooling is 14.5% for girls** (12% for boys). After 10 years of schooling, projected lifetime earnings roughly **double** relative to no schooling.
- **Marriage attenuates the protective association.** Education is strongly associated with lower NEET for unmarried girls, but the relationship is weaker among married girls — education and marriage prevention act as **complements**, not substitutes.
- **OLS and causal forest produce convergent estimates** in direction, magnitude, and stability. The investment case rests on triangulated evidence from two independent methods.

Section 6

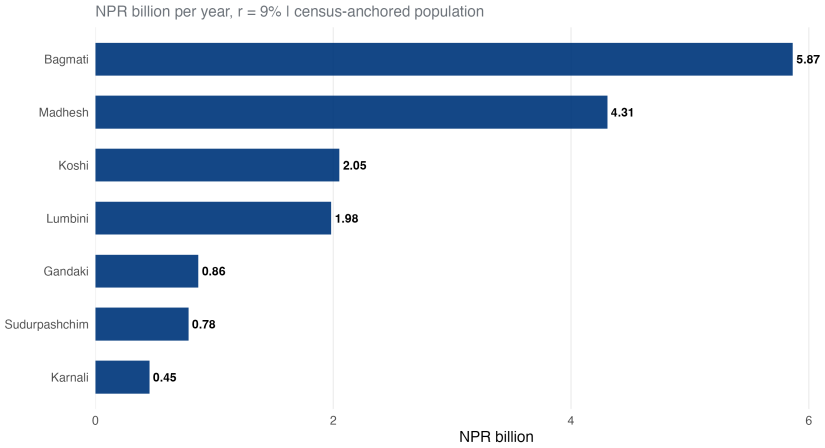
The Cost of Inaction

Nepal Forgoes NPR 16 Billion Every Year by Not Educating its Girls



Author's calculations, NLSS IV 2022/23 and Census 2021; MoF Red Book FY 2024/25.

Where the Cost Falls: Provincial Distribution



Author's calculations, NLSS IV 2022/23 and Census 2021.

What This Means in Concrete Policy Terms

- **NPR 16.3 billion (USD 123 million)** is the lifetime productivity loss from **one cohort** of 1,425,831 adolescent girls who will not complete secondary education. Counts are **census-anchored**: NLSS IV proportions applied to the 2021 Census female population (3,214,885 girls aged 15–24), because raw survey weights overstate population totals about fivefold.

- That equals about **18.9% of the national health-sector budget** (NPR 86.2B, FY2024/25), **15.4% of the social security allowance budget** (NPR 106B, FY2022/23), and **8% of the education-sector budget** (NPR 203.7B, FY2024/25).

- **Karnali, Sudurpashchim, and Madhesh** carry the largest relative burden — per-girl losses are smaller than in Bagmati because wages are lower, but a larger share of the local economy is at stake.

- **"Per cohort" means** the *lifetime* productivity loss for the current cohort of 1,425,831 adolescent girls (aged 15–24 today). A new cohort enters this stage every year, so this is a **recurring annual loss**, not one-off. Using the boys' 12% wage premium instead of the 9% ILO benchmark gives NPR 22.1 billion: the headline is the conservative floor.

- **Break-even spending per girl.** USD 123 million ÷ 1,425,831 girls **USD 86 per girl** in cumulative lifetime social spending. *Important caveat:* this break-even assumes every girl who completes secondary education enters the labour market and earns the median wage — which overstates realised returns, since many will not enter wage work. The figure is best read as the **theoretical maximum** spend that would still break even, not a prediction of per-girl returns.

Section 7

Recommendations

Four Priority Areas for Investment: Employment Opportunities First

1. **Livelihoods: pathways from education to employment.** Every gain in this report assumes jobs exist for newly qualified young women; the estimates are partial-equilibrium and depend on labour demand. Pair education with demand-side measures: skills bridges from upper-secondary to work, TVET, apprenticeships, second-chance and return-to-education pathways, and investment in employment-generating sectors. This is an explicit budget trade-off, to be planned together with education rather than in competition with it.
2. **Education: universal secondary completion with year-by-year retention.** Each year of schooling is associated with a 14.5% earnings premium for girls, and returns are *convex*, rising from ∼8% at primary to ∼27% at secondary and ∼35% at higher secondary. Programme design should reward *every year retained*, with the largest gains in upper-secondary.
3. **Marriage: treat prevention as an economic priority, starting in adolescence.** Marriage is associated with more variation in girls' NEET than poverty, caste, and disability combined; the married share rises from 9% at ages 15–18 to 57% at 19–24, and the penalty is severe in both groups. Dropout is itself a risk factor for child marriage, so retention and prevention reinforce each other. Enforcement, concentrated in Madhesh and Lumbini, must reach girls *before* the post-18 transition.
4. **Equity: close the gaps facing Dalit girls.** Education alone leaves Dalit women with a residual labour-market disadvantage that does not disappear when controlling for wealth. Pair education investment with apprenticeships, targeted scholarships, and anti-discrimination measures.

Reaching Those Already Left Behind

The four priorities need to reach three groups explicitly.

a. **Young people with disabilities.** Disability is associated with about **+23 pp NEET** among youth aged 15–24 and about **−20 pp enrolment** among girls (the enrolment estimate is imprecise). Include accessible schooling, reasonable accommodations and support for the transition into work. The analysis does not identify which disability-specific measures work best.

b. **Girls already married or out of school.** Married girls show very low enrolment and high NEET in every caste group and in both age groups. Prevention alone misses those already excluded: include flexible re-entry, second-chance learning and employment support. Childcare is an option worth assessing; it is not evaluated here.

c. **Madhesh and Lumbini.** Female NEET reaches about **65% in Madhesh and 50% in Lumbini**, the highest of the seven provinces. Prioritise both for integrated education-to-employment support, not only marriage enforcement. Madhesh also carries the second-largest cost of inaction (NPR 4.3 billion per cohort).

Section 8

Methodological Appendix

How We Estimate the Cost of Inaction

We combine three components:

1 · Causal effect
Generalised Random Forest (Wager & Athey 2018) on women 15–64, treatment = secondary completion, outcome = NEET. ATE = −13.2 pp.

2 · Earnings projection
Mincer wage equation on 952 women in long-term salaried employment. Conservative return = 9% per year of schooling (ILO benchmark, lower than the internal estimate of 14.5%).

3 · Population scaling
Census-anchored: NLSS IV weighted proportions applied to the Census 2021 female population (14.9M) give 3.21M girls aged 15–24, of whom 1.43M do not complete secondary. Provincial CoI = ATE × N excluded × PV per girl.

Discount rate: 5% | **Horizon**: 35 working years | **GDP source**: NSO Nepal 2023/24 provincial accounts.

Assumptions and Limitations

1. **Formal-sector selection bias.** Wages are observed only for the $\sim 25\%$ of women in wage employment. We test for this with a Heckman selection model (see appendix): selection is statistically significant ($\rho = 0.34$, $p < 0.01$) but correcting for it shifts the estimated return by less than 2 pp, so the Mincer estimates are robust within that specification; the selection test uses its own specification, whose uncorrected baseline (24.4%) differs from the main model's 14.5%. The conservative 9% ILO benchmark sits below all estimates.

2. **NEET-to-earnings transition.** The estimate assumes a girl who exits NEET earns the median wage *immediately*. In reality, job search, childcare, mobility, and social norms create friction that delays or reduces actual earnings. Assuming instant median-wage access therefore **overstates** the realised gain, making the cost-of-inaction figure an **upper bound** on what is realistically recoverable.

3. **Male migration not captured.** Boys leaving school for foreign employment generates remittances. The analysis does not yet quantify the parallel earnings that educated girls could generate through skilled migration.

4. **Population scaling.** Raw NLSS IV weights sum to 163.5M people against a census population of 29.2M, so absolute counts use survey proportions applied to the Census 2021 base. Proportions, regression estimates and per-girl results do not depend on this choice.

5. **Partial equilibrium.** Earnings gains are valued at current wages and current labour demand, and assume

The Mincer Wage Equation: Specification

The Mincer human-capital earnings function is estimated as:

$$\log(\text{wage}_{ip}) = \alpha + \beta_1 \cdot \text{schooling}_{ip} + \beta_2 \cdot \text{exp}_{ip} + \beta_3 \cdot \text{exp}^2_{ip} + \beta_4 \cdot \text{rural}_{ip} + \gamma_p + \varepsilon_{ip}$$

- $\log(\text{wage}_{ip})$ — natural log of monthly wage (NPR) for individual i in province p
- β_1 — **key parameter**: return per additional year of schooling (Mincer return)
- exp — years of potential labour market experience (age − schooling − 6)
- exp^2 — captures concavity (returns to experience diminish with age)
- rural — binary indicator for rural residence
- γ_p — province fixed effects (regional wage differentials)
- ε_{ip} — error term, heteroskedasticity-robust SE (HC1)

Specifications tested: (i) Basic — schooling, experience, rural; (ii) +Province FE; (iii) +Caste/ethnicity controls; (iv) Interactions with marital status. The reported estimate $\hat{\beta}_1 \approx 0.145$ implies a **14.5% return per year of schooling** for women; ILO Nepal benchmark of 9% used for the conservative CoI.

Appendix Table A1: Mincer Wage Equation, Female Salary Workers

Variable	Coefficient	SE	95% CI	p-value
Years of schooling	0.3298	0.0114	[0.3074, 0.3521]	<0.001
Experience (years)	0.1109	0.0108	[0.0898, 0.1320]	<0.001
Experience squared	-0.0014	0.0002	[-0.0019, -0.0009]	<0.001
Rural	-0.4351	0.0631	[-0.5588, -0.3114]	<0.001
N	2214.0000	NA	NA	NA
R-squared	0.3630	NA	NA	NA

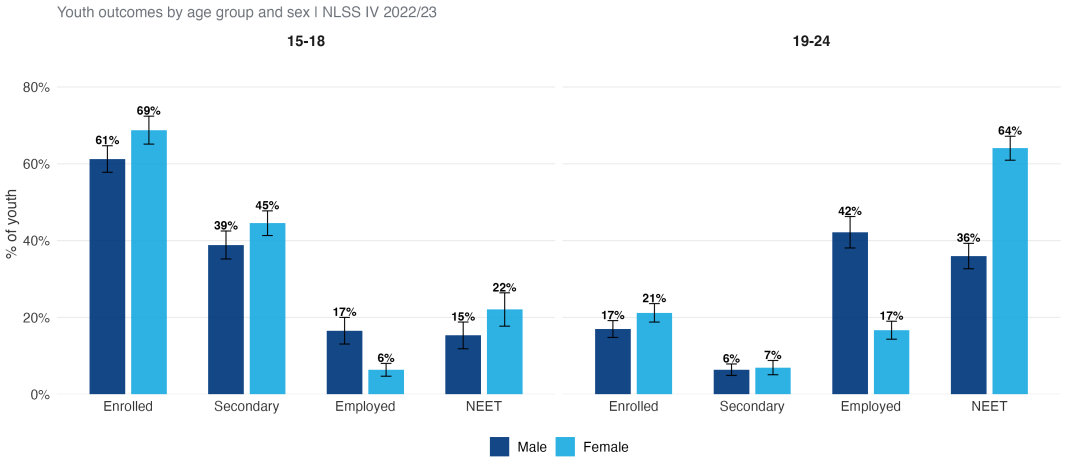
Notes: Dependent variable is log of monthly wage (NPR). Sample: women aged 15–64 in long-term salaried employment (N = 952). Specification includes years of schooling, experience, experience squared, rural indicator, and province fixed effects. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Appendix Table A2: Mincer Wage Equation, Male Salary Workers

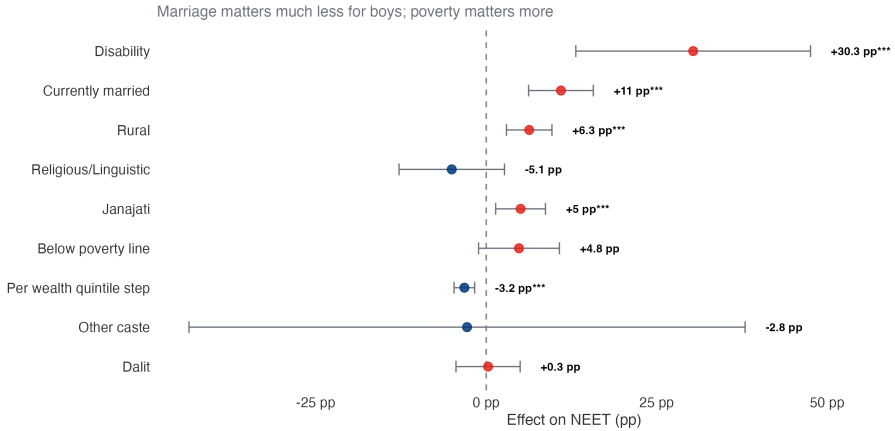
Variable	Coefficient	SE	95% CI	p-value
Years of schooling	0.1624	0.0065	[0.1497, 0.1751]	<0.001
Experience (years)	0.1193	0.0063	[0.1069, 0.1316]	<0.001
Experience squared	-0.0020	0.0001	[-0.0022, -0.0017]	<0.001
Rural	-0.3706	0.0359	[-0.4409, -0.3003]	<0.001
N	4677.0000	NA	NA	NA
R-squared	0.2290	NA	NA	NA

Notes: Dependent variable is log of monthly wage (NPR). Sample: men aged 15–64 in long-term salaried employment. Specification includes years of schooling, experience, experience squared, rural indicator, and province fixed effects. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Appendix Figure A1: Gender Gap by Age Group



Appendix Figure A2: Boys' NEET Regression — Coefficient Plot



LPM with province FE | HC1 SE | Ref: Brahmin/Chhetri, urban, non-poor.

Appendix Table A3: Boys' NEET Linear Probability Model

Variable	Coefficient	SE	95% CI	p-value
Rural	0.063	0.017	[0.030, 0.096]	<0.001
Below poverty line	0.048	0.030	[-0.011, 0.107]	0.112
Per wealth quintile step	-0.032	0.008	[-0.047, -0.017]	<0.001
Currently married	0.110	0.024	[0.062, 0.157]	<0.001
Disability	0.303	0.088	[0.131, 0.476]	<0.001
Janajati	0.050	0.019	[0.014, 0.087]	0.007
Dalit	0.003	0.024	[-0.044, 0.050]	0.911
Religious/Linguistic	-0.051	0.039	[-0.128, 0.027]	0.200
Other caste	-0.028	0.208	[-0.436, 0.380]	0.892

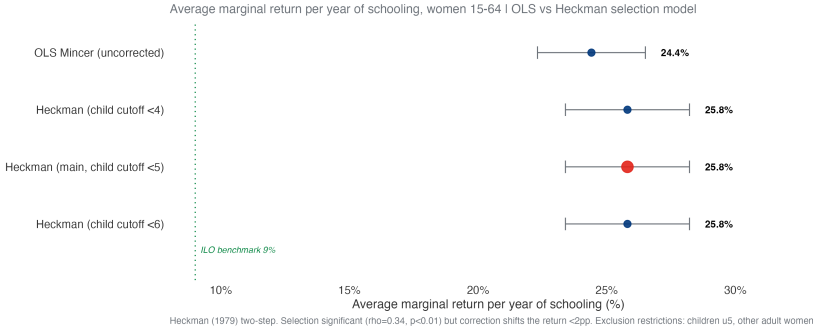
Notes: Dependent variable is NEET status. Linear probability model. Sample: boys aged 15–24 (n 4,300). Controls: rural, poverty status, wealth quintile, marital status, disability (WG 6+), caste/ethnicity. Reference categories: Brahmin/Chhetri, urban, non-poor. Province fixed effects included. Standard errors are heteroskedasticity-robust (HC1). *Source*: Author's calculations using NLSS IV 2022/23. *p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01.

Appendix: Heckman Selection Model

Wages are observed only for women in wage employment (about 25% of working-age women). If the women who work differ systematically from those who do not, the Mincer return could be biased. We test this with a Heckman (1979) two-step model.

- **Selection equation** (probit): probability of being in the wage sample, with exclusion restrictions — children under 5, other adult women in household, household size, and wealth quintile (factors affecting participation, not the wage itself).
- **Outcome equation**: the Mincer wage equation, corrected for selection via the inverse Mills ratio.
- **Result**: selection is statistically significant ($\rho = 0.34$, inverse Mills ratio $p < 0.01$) — there *is* positive selection into wage work. But the correction changes the estimated return by less than 2 pp.
- **Interpretation**: the Mincer estimates are robust to selection. This is *not* a causal-identification claim (that rests on the causal forest); it addresses only sample selection into observed wages.

Appendix: Returns are Robust Across Specifications



Average marginal return per year of schooling, women 15–64. The OLS (uncorrected, 24.4%) and Heckman-corrected (25.8%) estimates are nearly identical, and stable across the choice of child-age cutoff in the exclusion restriction. This test runs on its own specification; the 14.5% headline return comes from the main Mincer model, so the test speaks to stability under selection, not to the level of the headline. The 9% ILO benchmark used for the cost-of-inaction calculation is far below all estimates — making the CoI a conservative figure. Returns are convex: marginal return rises from ~8% at primary to ~27% at secondary and ~35% at higher secondary.

Thank you

hgrueso@unicef.org